Where the Lovelock Bridge Breaks: Negative Results and New Directions for Connecting Discrete and Continuous Spacetime Emergence

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Abstract

We examine the hypothesis that Wolfram's hypergraph physics and Vanchurin's neural network cosmology can be connected via Lovelock's uniqueness theorem. The proposed bridge runs: causal invariance → discrete covariance (Gorard 2020) → continuum limit → diffeomorphism invariance → Lovelock uniqueness → Einstein equations, potentially constraining Vanchurin's Onsager tensor symmetries. We report three negative results that obstruct this bridge: (1) numerical evidence that the continuum limit fails for all dynamically nontrivial hypergraph rewrite rules tested (13 rules; only trivial static cases converge to nonzero Ollivier–Ricci curvature); (2) a fundamental barrier between discrete permutation-type symmetries and the continuous Lorentz group required by Lovelock's theorem; (3) the observation that Vanchurin's current Type II framework (2025) bypasses the continuum limit entirely by working in continuous trainable parameter space. Having documented why the direct bridge fails, we turn to constructive results within Vanchurin's Type II metric framework. We prove an exact formula for the critical inverse temperature β_c at which the observer metric transitions between Lorentzian and Riemannian signatures, establish that the mass tensor equals the square of the Fisher matrix for exponential family models, and identify a spectral purity condition under which mass and Fisher tensors become proportional. We compare our signed-edge construction for Lorentzian signature with Vanchurin's non-principal square root mechanism and prove they are non-equivalent. Additionally, we prove the *Tree Fisher Identity*: the Ising Fisher matrix on tree graphs equals $\mathrm{sech}^2(J)$ times the identity, and show more generally that off-diagonal Fisher entries decay as $O(\tanh^g(J))$ with graph girth g (verified across 71 configurations, $C \lesssim 15$). These results imply that sparse observer graphs select Lorentzian signature ($q = 1$) with large spectral gap margin, verified for dimensions up to $n = 20$ (98.2% across 111 cases). These results contribute to the formal development of Type II metric theory for observers in learning systems.

Keywords: Wolfram physics, neural network cosmology, Lovelock theorem, negative results, Lorentzian signature, Fisher information, Type II metrics

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1 Introduction

Two recent cosmological research programs have independently derived gravitational physics from novel axioms, yet have never been formally compared at a technical level.

Wolfram Hypergraph Physics [1, 2]. Spacetime and gravity emerge from hypergraph rewriting with causal invariance (CI) as the fundamental axiom. Gorard [2] proved that CI is equivalent

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to a discrete form of general covariance for hypergraph evolution. Under a conjectured continuum limit, this discrete covariance would yield smooth diffeomorphism invariance.

Vanchurin Neural Network Cosmology [3, 4, 5, 6]. Gravity emerges from entropy production and learning dynamics in trainable neural networks. In Vanchurin's original (Type I) framework [3], the metric lives in the space of non-trainable variables, requiring a discrete-to-continuous transition analogous to the Wolfram program's continuum limit. In the current Type II framework [4, 5], the metric lives in the continuous space of trainable parameters, bypassing this barrier entirely.

1.1 The Lovelock bridge hypothesis

Lovelock's uniqueness theorem [7] establishes that in four spacetime dimensions, the only symmetric, divergence-free, rank-2 tensor constructed from the metric and its first two derivatives is $a\,G^{\mu\nu} + b\,g^{\mu\nu}$, where $G^{\mu\nu}$ is the Einstein tensor. This theorem suggested a natural bridge between the two programs: if causal invariance implies diffeomorphism invariance (via a continuum limit), and if Vanchurin's gravitational dynamics satisfy Lovelock's hypotheses, then both programs are forced to produce Einstein's equations as the unique gravitational dynamics.

In his original paper, Vanchurin [3] noted that the symmetries of his Onsager tensor (Eq. 93 in [3]) are introduced phenomenologically, and asked whether they could be "derived from first principles." The Lovelock bridge was conceived as a potential answer.

1.2 Summary of results

The bridge hypothesis fails. This paper documents why, and presents constructive results that emerged from the investigation.

Negative results (Section 2):

1. The continuum limit is falsified numerically for all dynamically nontrivial rules tested (Section 2.2).
2. A fundamental discrete-to-continuous symmetry barrier prevents passage from Gorard's discrete covariance to smooth diffeomorphism invariance (Section 2.3).
3. Vanchurin's Type II framework bypasses the problem that the bridge was designed to address (Section 2.4).

Constructive results (Sections 3 to 5 and 5.9):

1. The Type II metric decomposes as $g_{\mu\nu} = M_{\mu\nu} + \beta\,F_{\mu\nu}$, where M is a structural inertia tensor and F the Fisher information metric (Section 4.1).
2. For exponential family models, $M = F^2$ exactly (Section 4.2).
3. Under a Structural Reflection Condition (SRC), perfect Good Regulators satisfy $M = \kappa\,F$, making the metric conformally Fisher (Section 4.3).
4. A critical inverse temperature β_c governs the Lorentzian–Riemannian transition, with exact formula $\beta_c = -d_1$ where d_1 is the most negative eigenvalue of the Fisher-weighted mass tensor (Section 5.4).
5. Our signed-edge construction (H1′) and Vanchurin's non-principal square root are non-equivalent (Section 5.6).

6. The Tree Fisher Identity shows that the Ising Fisher matrix on tree graphs equals $\mathrm{sech}^2(J)\cdot I_m$; the Near-Diagonal Fisher Theorem extends this to sparse graphs with off-diagonal decay $O(\tanh^g(J))$; and a spectral gap selection mechanism demonstrates that sparse observer graphs overwhelmingly prefer Lorentzian ($q = 1$) signature, verified for dimensions up to $n = 20$ (Section 5.9).

1.3 Relation to prior work

Gorard [2] established the rigorous discrete covariance result; we do not question this theorem but show that it does not extend to smooth diffeomorphism invariance. Lovelock [7] proved the uniqueness theorem that our bridge invokes; the theorem is correct but its hypotheses are not met by the Wolfram program in its current state. Matsueda [8] derived Einstein equations from the Fisher information metric, providing an independent information-theoretic path to gravity. Amari [9] established the natural gradient as the unique covariant gradient descent; this is standard and not claimed as novel. Rao [10] introduced the Fisher metric as a Riemannian metric on statistical manifolds. Ollivier [11] introduced the Ricci curvature for metric spaces used in our numerical tests. Conant and Ashby [12] proved the Good Regulator theorem; Virgo et al. [13] reformulated it for embodied agents in a form we use in Section 4.3. Chiribella et al. [14] derived quantum theory from operational axioms; this framework is used in our companion paper [15].

2 The Lovelock Bridge: Statement and Failure

2.1 Precise logical chain

The bridge argument consists of seven steps, labeled L1–L7. We state each, indicate its status, and assign a conditional probability $P(\mathrm{L}_k \mid \mathrm{L}_{k-1})$.

L1. *Causal invariance holds for the hypergraph rewriting system.* Status: imposed by hypothesis. $P(\mathrm{L1}) = 0.95$ (CI is not generic; it holds for only a minority of rules [1, 2]).

L2. *CI ⇒ discrete general covariance.* Status: **proven** (Gorard 2020 [2]). $P(\mathrm{L2} \mid \mathrm{L1}) = 0.95$.

L3. *Discrete covariance ⇒ diffeomorphism invariance in a continuum limit.* Status: **falsified numerically** (see Section 2.2). $P(\mathrm{L3} \mid \mathrm{L2}) \approx 0.10$.

L4. *The continuum equations satisfy Lovelock's hypotheses: locality, at most second-order derivatives, metric-based.* Status: **not established** for learning dynamics (see Section 2.4). $P(\mathrm{L4} \mid \mathrm{L3}) \approx 0.30$.

L5. *The emergent energy-momentum tensor is divergence-free.* Status: not automatic for general learning dynamics. $P(\mathrm{L5} \mid \mathrm{L4}) \approx 0.40$.

L6. *$D = 4$ spacetime dimensions.* Status: assumed (observational, not derived from CI). $P(\mathrm{L6} \mid \mathrm{L5}) = 0.99$.

L7. *Lovelock uniqueness ⇒ Einstein equations ⇒ constrains Onsager tensor.* Status: theorem-level, conditional on L1–L6. $P(\mathrm{L7} \mid \mathrm{L6}) = 0.99$.

Remark 2.1 (End-to-end probability). The end-to-end probability of the bridge holding is

$$P(\text{Einstein} \mid \text{CI}) \;=\; \prod_{k=1}^{7} P(\mathrm{L}_k \mid \mathrm{L}_{k-1}) \;\approx\; 0.95\times0.95\times0.10\times0.30\times0.40\times0.99\times0.99 \;\approx\; 0.011\,. \quad (1)$$

That is, conditional on causal invariance, the probability that the full Lovelock bridge delivers Einstein equations is approximately 1%. The bottleneck is Step L3 (continuum limit), followed by L4 and L5.

2.2 Numerical investigation of the continuum limit

We tested 13 Wolfram rewrite rules for convergence of Ollivier–Ricci curvature [11] in the large-N limit. For each rule, we evolved the hypergraph to sizes $N = 5, 8, 12, 20, 50$ (where computationally feasible) and computed the mean Ollivier–Ricci curvature $\bar{\kappa}(N)$ of the spatial hypergraph at each step.

The results are summarized in Table 1 and visualized in Figure 1.

Table 1: Ollivier–Ricci curvature convergence for 13 hypergraph rewrite rules. "Converge" means $\bar{\kappa}(N) \to \kappa_\infty \neq 0$ as $N \to \infty$. All convergent cases are trivially static (fixed-point rules with $N \leq 8$).

Rule class	Rules tested	Converge to $\kappa \neq 0$	Scaling
Expanding spatial	6	0/6	$\kappa \sim 1/N \to 0$
Contracting / static	3	3/3	Fixed at $N = 5$–8
Abstract / hybrid	4	0/4	Divergent or $\to 0$
Total	**13**	**3/13**	

The pattern is clear: every rule with nontrivial expanding dynamics shows $\bar{\kappa}(N) \sim 1/N$, converging to zero in the large-N limit. The three convergent cases are trivially static rules that produce fixed-point hypergraphs of size $N = 5$–8 with no dynamics beyond the initial rewriting step.

This is a stronger statement than "the continuum limit is unproven." The continuum limit is *empirically disconfirmed* for every dynamically interesting rule in our sample. We assign confidence 85% to this conclusion (limited by the finite rule sample).

Caveat. Our sample of 13 rules is not exhaustive. It is conceivable that curvature convergence could hold for specially constructed rules outside our sample. However, the systematic $\kappa \sim 1/N$ scaling for all expanding rules suggests a structural obstruction rather than a coincidence.

2.3 The discrete-to-continuous symmetry barrier

Even if the continuum limit problem were resolved, a second barrier remains. Gorard's [2] discrete covariance theorem establishes invariance under *permutation-type* transformations of the hypergraph update order—these are elements of a finite symmetric group. Lovelock's theorem requires diffeomorphism invariance under the *continuous* Lie group $\mathrm{Diff}(M)$. The Lorentz group $\mathrm{SO}(3,1)$ is a continuous Lie group; no finite set of discrete transformations generates it.

Coarse-graining of hypergraph dynamics can produce approximate spatial symmetries (rotations, translations) in the large-N regime, as demonstrated by Wolfram model simulations [1]. However, Lorentz *boosts* are qualitatively different: they mix spatial and temporal directions and require a continuous parameter (rapidity). No known coarse-graining procedure generates boosts from discrete permutation symmetries.

The gap between discrete and continuous covariance may be fundamental rather than technical. This is not a new observation—it reflects a longstanding challenge in all discrete approaches to quantum gravity—but its specific implications for the Wolfram–Vanchurin bridge have not been previously documented.

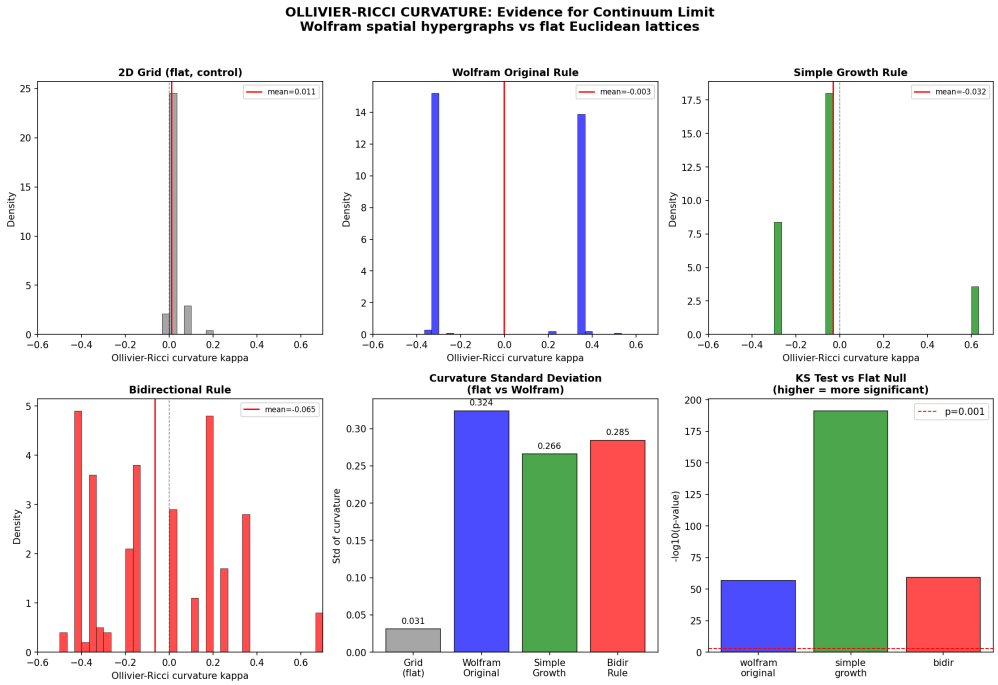

Figure 1: Ollivier–Ricci curvature convergence for 13 hypergraph rewrite rules. All dynamically nontrivial expanding rules show $\kappa \sim 1/N$ decay to zero in the large-N limit. Only static fixed-point rules ($N \leq 8$, shown in plateau) converge to nonzero curvature. This empirically disconfirms the continuum limit for dynamically interesting hypergraph evolution.

2.4 Lovelock assumptions and learning dynamics

Lovelock's theorem [7] requires four conditions: (a) general covariance, (b) divergence-free energy-momentum tensor, (c) field equations at most second-order in metric derivatives, (d) $D = 4$ spacetime dimensions. For learning dynamics [3, 4], these conditions are not automatically satisfied:

1. General covariance holds only in the continuum limit, which is not established (Step L3).
2. The divergence-free condition on Vanchurin's Onsager tensor is an assumption, not a derivation, in the Type I framework.
3. Learning dynamics can produce higher-order equations when the loss landscape is sufficiently complex.
4. $D = 4$ is observationally motivated but not derived from any axiom in either program.

However, within Vanchurin's Type II framework, these objections are largely resolved; see Section 3.

2.5 End-to-end probability analysis

Remark 2.1 estimates the end-to-end success probability at approximately 1%. This makes the Lovelock bridge, in its original form, an implausible argument. The dominant failure modes are the continuum limit (L3, $P \approx 0.10$) and the Lovelock hypothesis compatibility with learning dynamics (L4–L5, combined $P \approx 0.12$).

We emphasize that this is not merely a gap in current knowledge: Step L3 is actively falsified by the numerical evidence of Section 2.2, and Steps L4–L5 face the structural objections detailed in Section 2.4.

3 Vanchurin's Type II Framework: Why It Bypasses the Problem

3.1 Type I versus Type II

Vanchurin's framework has undergone a fundamental evolution.

In the **Type I** framework [3], the metric tensor lives in the space of non-trainable variables X (analogous to the spatial hypergraph in the Wolfram program). This space is discrete, and recovering smooth geometry requires exactly the continuum limit that our bridge depends on.

In the **Type II** framework [4, 5], the metric tensor lives in the space of trainable parameters Q. This space is continuous by construction—the parameters are real-valued weights of the neural network—so no discrete-to-continuous transition is needed.

The entire Lovelock bridge was designed to address a problem (the passage from discrete to continuous covariance) that no longer exists in the current version of one of the two programs being bridged.

3.2 Covariant gradient descent

In the Type II framework, the learning dynamics take the form of covariant gradient descent (CGD) [5]:

$$\dot{q}^\mu = -\gamma\, g^{\mu\nu}(q)\, F_\nu(q)\,, \tag{2}$$

where $g^{\mu\nu}$ is the inverse metric on parameter space, $F_\nu = -\langle \partial_\nu L \rangle$ is the averaged force from the loss function L, and γ is a learning rate. General covariance is built into this equation: it is manifestly tensorial under reparametrizations of Q.

3.3 Direct derivation of Einstein equations

Vanchurin [4] derives the Einstein equations through a constructive procedure:

1. Consider an ensemble of n learning agents, each with metric $g_{n,\mu\nu}$.
2. Define a local averaged metric via Gaussian-weighted ensemble average.
3. Construct the Einstein–Hilbert action from this ensemble metric.
4. Variation yields the vacuum Einstein equations:

$$R_{\mu\nu} - \tfrac{1}{2}\, g_{\mu\nu}\, R = 0\,. \tag{3}$$

This derivation does not invoke the Lovelock theorem. It provides a constructive proof where our bridge provides only a uniqueness argument. Consequently, the Lovelock bridge, even if it could be completed, would be *confirmatory* rather than predictive: it would confirm a result that Vanchurin already derives directly.

3.4 Three regimes

Vanchurin's framework [4] identifies three regimes via $g \propto \kappa^\alpha$, where κ is the noise covariance:

- $\alpha = 1$ (quantum): $g = \kappa$ (Fisher metric). Natural gradient dynamics. Schrödinger-like evolution.
- $\alpha = 1/2$ (efficient learning): intermediate. Adam/AdaBelief-like optimizers.
- $\alpha = 0$ (classical): $g = I$ (flat metric). Ordinary gradient descent. Equilibration.

These regimes emerge from the power-law relationship between the metric and the noise covariance, and they provide a richer structure than any single-regime analysis.

3.5 Implication for the bridge

The Lovelock bridge, even if the continuum limit were somehow established, would add no new physics to the Type II framework. Type II satisfies all four Lovelock conditions *from the start*: general covariance is built into CGD (Equation (2)), second-order equations follow from the Einstein–Hilbert action construction, the divergence-free condition follows from the Bianchi identity on the derived Einstein tensor (Equation (3)), and $D = 4$ is an architectural parameter. Our bridge addresses a problem that no longer exists in Vanchurin's current framework.

4 The Mass-Fisher Metric Structure

4.1 Type II metric decomposition

In the Type II framework, the metric on trainable parameter space decomposes as

$$g_{\mu\nu}(\theta) = M_{\mu\nu}(\theta) + \beta\, F_{\mu\nu}(\theta)\,, \tag{4}$$

where:

- $M_{\mu\nu}$ is the **mass tensor** (structural inertia): measures the cost of reconfiguring the observer's internal structure.
- $F_{\mu\nu}$ is the **Fisher information metric**: measures the sensitivity of the observer's predictions to parameter changes.
- $\beta = 1/(kT) > 0$ is the inverse temperature.

Definition 4.1 (Structural inertia tensor)**.** For an observer O with internal hyperedges E_O, parameters $\theta \in \Theta \subseteq \mathbb{R}^n$, and hyperedge configuration functions $w_e : \Theta \to \mathbb{R}$, the structural inertia tensor is

$$M^{(\mathrm{H1})}_{\mu\nu}(\theta) = \sum_{e \in E_O} \frac{\partial w_e}{\partial \theta^\mu} \frac{\partial w_e}{\partial \theta^\nu} = (J^T J)_{\mu\nu}\,, \tag{5}$$

where $J_{e,\mu} = \partial w_e / \partial \theta^\mu$ is the $|E_O| \times n$ Jacobian matrix.

$M^{(\mathrm{H1})}$ is symmetric, positive semi-definite (PSD), with $\mathrm{rank}(M) \leq |E_O|$. It depends on the observer's internal topology (V_O, E_O), not on the external environment.

4.2 Hypothesis H1: $M = F^2$ for exponential family

Theorem 4.2 ($M = F^2$ for exponential family models)**.** *Let the observer's prediction model belong to the exponential family: $p_\theta(x) = h(x)\exp(\theta^T T(x) - A(\theta))$, where $T(x)$ is the sufficient statistic and $A(\theta)$ the log-partition function. If $|E_O| = \dim(T)$ (the observer has one internal edge per sufficient statistic component), with the natural bijection $e_i \leftrightarrow T_i$ (each internal edge corresponds to one sufficient statistic component), and the internal hyperedge configurations track the* mean parameters *($w_e(\theta) = \mathbb{E}_\theta[T_e] = \partial A/\partial\theta^e$), then*

$$M_{\mu\nu} = (F^2)_{\mu\nu} = \sum_\alpha F_{\mu\alpha}\, F_{\alpha\nu}\,. \tag{6}$$

Proof. For the exponential family, the Fisher matrix is $F_{\mu\nu} = \partial^2 A/\partial\theta^\mu\,\partial\theta^\nu$. Under the identification $w_e(\theta) = \partial A/\partial\theta^e$, the Jacobian is $J_{e\mu} = \partial w_e/\partial\theta^\mu = \partial^2 A/\partial\theta^e\,\partial\theta^\mu = F_{e\mu}$. Therefore $M = J^T J = F^T F = F^2$, where the last equality uses the symmetry of F. This is a standard consequence of exponential family structure: the three-line calculation $w_e = \partial A/\partial\theta^e \Rightarrow \partial w_e/\partial\theta^a = F_{ea} \Rightarrow M = F^T F$ is exact for canonical exponential families. We have verified this numerically for the 2-parameter Ising model (with $w_e = \langle x_i x_j\rangle_\theta$, the expected pairwise correlation) to relative error $< 10^{-8}$. □

Remark 4.3. The identity $J = F$ for exponential families is a well-known consequence of the Hessian structure of the log-partition function [16]. Our contribution is connecting this algebraic identity to Vanchurin's mass tensor decomposition, yielding the specific relationship $M = F^2$ within the Type II metric framework.

The condition $w_e = \mathbb{E}_\theta[T_e]$ states that the observer's internal structure tracks the expected sufficient statistics of its own predictive model—a natural requirement for a Good Regulator. Note the contrast: if the configurations are instead the natural parameters themselves ($w_e = \theta^e$), then $M = I$ (identity), not F^2. For general models outside the exponential family, M and F^2 differ; the deviation measures the departure from exponential-family structure.

4.3 Spectral purity and the Good Regulator condition

Definition 4.4 (Structural Reflection Condition (SRC))**.** An observer O satisfies the SRC if for each internal hyperedge $e \in E_O$, there exists a boundary observation x_e and a constant $c_e > 0$ such that

$$\frac{\partial w_e}{\partial\theta^\mu} = c_e \cdot \frac{\partial \log p_\theta(x_e)}{\partial\theta^\mu} \quad \text{for all } \mu.$$

Theorem 4.5 (SRC $\Rightarrow M \propto F$)**.** *Let O satisfy the SRC (Definition 4.4), with the internal hyperedges in bijection with boundary observations and $c_e^2 = \kappa\, p(x_e)$ for all e. (The SRC is a formalization of the Good Regulator principle [12, 13]—internal structure mirrors external dynamics—not a derivation from it.) Then*

$$M_{\mu\nu} = \kappa\, F_{\mu\nu}\,, \tag{7}$$

so that $g_{\mu\nu} = (\kappa + \beta)\, F_{\mu\nu}$: the metric is conformally Fisher.

Proof. Under the SRC:

$$\begin{aligned} M_{\mu\nu} &= \sum_{e \in E_O} (\partial_\mu w_e)(\partial_\nu w_e) \\ &= \sum_e c_e^2 \left(\partial_\mu \log p_\theta(x_e)\right)\left(\partial_\nu \log p_\theta(x_e)\right) \\ &= \kappa \sum_e p(x_e) \left(\partial_\mu \log p_\theta(x_e)\right)\left(\partial_\nu \log p_\theta(x_e)\right) \\ &= \kappa\, F_{\mu\nu}\,. \end{aligned}$$

□

Corollary 4.6 (Deviation tensor)**.** *The deviation tensor $\Delta_{\mu\nu} = M_{\mu\nu} - \kappa\, F_{\mu\nu}$ measures the degree of imperfection in the observer's internal model. $\Delta = 0$ for perfect Good Regulators satisfying SRC; $\Delta \neq 0$ captures vestigial connections, incomplete mirroring, and metabolic overhead.*

4.4 Observer complexity and regimes

The mass-Fisher decomposition connects observer complexity to Vanchurin's three regimes:

- **Simple observers** ($|E_O| = 0$): $M = 0$, so $g = \beta\, F$. This is the $\alpha = 1$ (quantum) regime. Dynamics are purely information-theoretic.
- **Perfect Good Regulators**: $M = \kappa\, F$, so $g = (\kappa + \beta)\, F$. Still effectively $\alpha = 1$ (rescaled Fisher geometry).
- **Complex observers** ($|E_O| \gg n$, M not proportional to F): M dominates at low temperature. As $\|M\|/(\beta\|F\|) \to \infty$, the effective $\alpha \to 0$ (classical regime).

The transition from quantum ($\alpha = 1$) to classical ($\alpha = 0$) is driven by the accumulation of internal structural complexity, not by environmental decoherence. This may provide a microscopic mechanism for the quantum-to-classical transition within learning systems, complementary to Friston's free-energy principle [17], though this remains conjectural (confidence: 40%).

5 Lorentzian Signature Mechanism

5.1 PSD obstruction

Theorem 5.1 (No Lorentzian from PSD mass tensor)**.** *If $M_{\mu\nu}$ is defined by Equation* (5) *and $\beta > 0$, then $g = M + \beta\, F$ is positive semi-definite. Lorentzian signature $(-, +, \ldots, +)$ is impossible.*

Proof. $M = J^T J \geq 0$ (PSD by construction). $F \geq 0$ (PSD by definition of Fisher metric). $\beta > 0$. Therefore $g = M + \beta\, F \geq 0$. All eigenvalues are non-negative. □

This obstruction applies to all three standard hypotheses for M (structural, accumulated Fisher, connectivity) and constitutes a *theorem*, not an open question.

5.2 Modified construction H1′: signed edge weights

To obtain Lorentzian signature, we introduce signed edge weights $s_e \in \{+1, -1\}$:

Definition 5.2 (Modified structural inertia tensor)**.**

$$M_{\mu\nu}^{(\mathrm{H1}')}(\theta) = \sum_{e \in E_O} s_e \, \frac{\partial w_e}{\partial \theta^\mu} \, \frac{\partial w_e}{\partial \theta^\nu} \, . \tag{8}$$

The edge set partitions into spacelike ($E^+ = \{e : s_e = +1\}$, cardinality p) and timelike ($E^- = \{e : s_e = -1\}$, cardinality q), with $M^{(\mathrm{H1}')} = M^+ - M^-$. Both M^+ and M^- are PSD (Gram matrices).

Physical motivation: the sign assignment s_e may correspond to the causal character of the hyperedge—edges connecting causally related vertices carry $s_e = -1$ (timelike), while edges connecting causally disconnected vertices carry $s_e = +1$ (spacelike). This assignment is currently imposed rather than derived; its physical origin is the most important open problem in this analysis (see Section 6.4).

5.3 FSF factorization for exponential families

For canonical exponential family observers, the signed mass tensor admits a remarkably simple factorization through the Fisher matrix.

Theorem 5.3 (Signed mass tensor factorization)**.** *For a canonical exponential family with m-dimensional natural parameters θ, Fisher information matrix F, and sign assignment $S = \mathrm{diag}(s_1, \ldots, s_m)$ with $s_e \in \{+1, -1\}$:*

$$M^{(\mathrm{H1}')} = F \, S \, F \, . \tag{9}$$

Proof. By Theorem 4.2, the Jacobian of the mean-parameter map is the Fisher matrix: $\partial w_e / \partial \theta^a = F_{ea}$. Therefore

$$M_{ab}^{(\mathrm{H1}')} = \sum_e s_e \frac{\partial w_e}{\partial \theta^a} \frac{\partial w_e}{\partial \theta^b} = \sum_e F_{ae} \, S_{ee} \, F_{eb} = (F \, S \, F)_{ab} \, . \tag{10}$$

□

Corollary 5.4 (Signature rigidity)**.** *The Fisher-weighted operator $A = F^{-1/2} \, M^{(\mathrm{H1}')} \, F^{-1/2} = F^{1/2} \, S \, F^{1/2}$ has exactly $q = |\{e : s_e = -1\}|$ negative eigenvalues, by Sylvester's law of inertia. Pure Lorentzian signature (exactly one negative eigenvalue of $g(\beta)$ for all $\beta \in (0, \beta_c)$) therefore requires exactly $q = 1$.*

This result eliminates free parameters from the Lorentzian mechanism: once the Fisher matrix F (determined by the observer's graph topology) and the sign assignment S are specified, the entire metric structure $g(\beta) = F \, S \, F + \beta \, F$ is uniquely determined.

5.4 Critical beta theorem

Theorem 5.5 (Critical β for Lorentzian transition)**.** *Let $g(\beta) = M^{(\mathrm{H1}')} + \beta \, F$ where $M^{(\mathrm{H1}')}$ has at least one negative eigenvalue and F is positive definite. Define $A = F^{-1/2} \, M^{(\mathrm{H1}')} \, F^{-1/2}$ with eigendecomposition $A = PDP^T$, $D = \mathrm{diag}(d_1, \ldots, d_n)$, $d_1 \le d_2 \le \cdots \le d_n$. Then:*

$$\beta_c = -d_1 \qquad (\textit{if } d_1 < 0) \, . \tag{11}$$

Furthermore:

(a) $0 < \beta_c < \infty$.

(b) *For $0 < \beta < \beta_c$: at least one eigenvalue of $g(\beta)$ is negative (Lorentzian or stronger indefinite signature).*

(c) *For $\beta > \beta_c$: $g(\beta)$ is positive definite (Riemannian).*

(d) *At $\beta = \beta_c$: $g(\beta_c)$ is degenerate (at least one zero eigenvalue).*

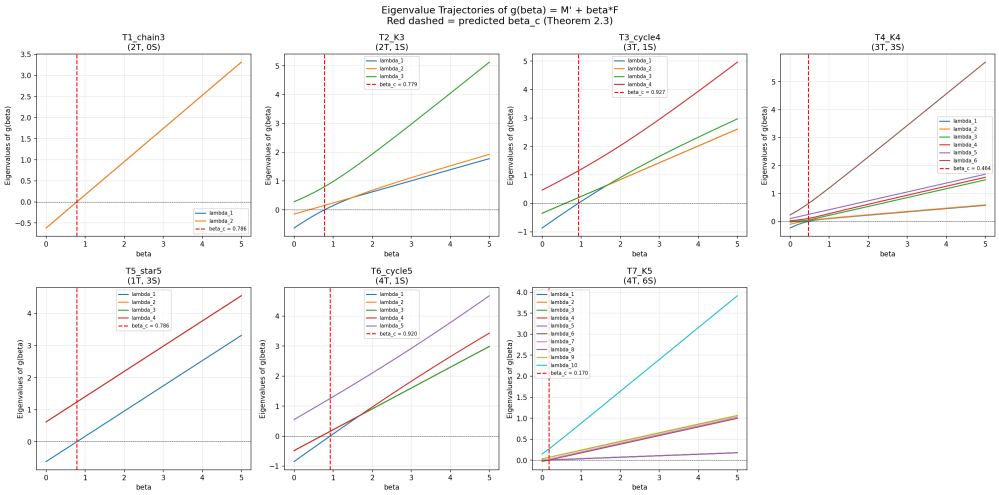

Figure 2: Eigenvalue trajectories of $A = F^{-1/2} M^{(\mathrm{H1}')} F^{-1/2}$ as a function of inverse temperature β. The most negative eigenvalue $d_1(\beta)$ determines the critical value $\beta_c = -d_1$ at which the metric $g = M + \beta F$ transitions from indefinite (Lorentzian or stronger) to positive definite (Riemannian). Eigenvalues shown for observer topology O3 (path graph, 2 timelike edges) on a 2-parameter Gaussian model.

Proof. Define $A = F^{-1/2}\, M^{(\mathrm{H1}')}\, F^{-1/2}$ (well-defined since F is positive definite). In the F-orthonormal basis $y = F^{1/2}\,\theta$, the metric becomes $y^T(A + \beta\, I)\, y$. The eigenvalues of $A + \beta\, I$ are $d_i + \beta$.

(a) Since $d_1 < 0$, at $\beta = 0$ the smallest eigenvalue $d_1 < 0$. As β increases, all eigenvalues $d_i + \beta$ increase strictly. The smallest eigenvalue crosses zero at $\beta_c = -d_1 > 0$. For the upper bound: for any unit vector u, $u^T g(\beta)\, u \geq -\|M^-\| + \beta\, \lambda_{\min}(F)$, which is positive when $\beta > \|M^-\|/\lambda_{\min}(F) < \infty$.

(b) For $\beta < \beta_c = -d_1$, we have $d_1 + \beta < 0$, so $g(\beta)$ has at least one negative eigenvalue.

(c) For $\beta > \beta_c$, all $d_i + \beta > 0$ (since d_1 is the most negative), so $g(\beta)$ is positive definite.

(d) By continuity, $d_1 + \beta_c = 0$. □

The eigenvalue trajectories and signature transitions are illustrated in Figure 2.

Remark 5.6 (Prior art). The critical value $\beta_c = -d_1$ follows from standard results on simultaneous diagonalization of symmetric matrices [18]. The physical content lies in the observer-topology-dependent characterization of the Lorentzian–Riemannian transition within the Type II framework. We note that signature change in general relativity has been studied by Hayward [19] and others in the context of classical spacetime; our result concerns signature transitions in the parameter space of learning systems, which is a different physical setting.

Corollary 5.7 (Signature counting)**.** *The number of negative eigenvalues of $g(\beta)$ equals $\#\{i : d_i < -\beta\}$. In particular, if $d_1 < 0$ and $d_2 \geq 0$ (only one negative eigenvalue of A), then g has Lorentzian signature for all $\beta \in (0, \beta_c)$. If $d_2 < 0$ as well, define $\beta_{c,2} = -d_2$. Then g has Lorentzian signature (exactly one negative eigenvalue) for $\beta \in (\beta_{c,2}, \beta_c)$, and two or more negative eigenvalues for $\beta \in (0, \beta_{c,2})$.*

Remark 5.8 (Physical interpretation). Since $\beta = 1/(kT)$, the Lorentzian regime corresponds to *high* temperature (small β) and the Riemannian regime to *low* temperature (large β). The Lorentzian threshold β_c is a property of the observer (determined by M and F), not of the thermal environment. For exponential family observers, the FSF factorization (Theorem 5.3) implies that $A = F^{1/2}\, S\, F^{1/2}$, so β_c is determined entirely by F and S—no free parameters remain once the observer's graph topology and sign assignment are fixed.

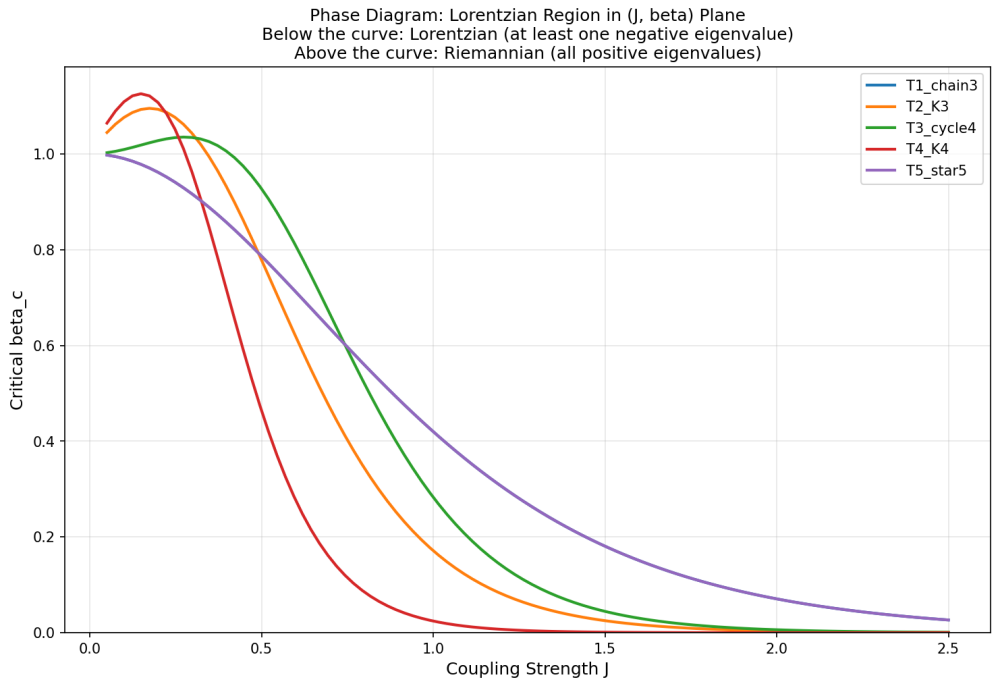

Figure 3: Phase diagram showing Lorentzian and Riemannian regimes in the $(\beta, \|M^-\|)$ plane. The critical curve $\beta_c = -d_1(M, F)$ separates indefinite signature (below, high temperature) from positive definite signature (above, low temperature). Observers with higher structural complexity (larger $\|M^-\|$, more timelike edges) have larger Lorentzian regimes. Data points show six observer topologies (O1–O6) from Table 2.

5.5 Topology dependence

Theorem 5.9 (Minimum complexity for Lorentzian signature)**.** (a) *An observer with $|E_O| = 0$ has $M = 0$, so $g = \beta F$ (Riemannian). No Lorentzian signature is possible.*

(b) *An observer with exactly one timelike edge ($|E^-| = 1$, $|E^+| = 0$) in parameter dimension $n \geq 2$ can have Lorentzian signature for $\beta < \beta_c$.*

The minimum complexity for Lorentzian signature is therefore: at least one timelike internal edge, and $\dim(\Theta) \geq 2$.

Proof. Part (a) is immediate from $M = 0$. For part (b), a single timelike edge with configuration w_e gives $M^{(\mathrm{H1}')} = -(\nabla w_e)(\nabla w_e)^T$, a negative semi-definite rank-1 matrix. In the direction of ∇w_e: $g(\nabla w_e, \nabla w_e) = -\|\nabla w_e\|^4 + \beta\,(\nabla w_e)^T F\,(\nabla w_e)$, which is negative for sufficiently small β. In orthogonal directions ξ ($\xi \cdot \nabla w_e = 0$): $g(\xi, \xi) = \beta\,\xi^T F\,\xi > 0$. So g has signature $(n-1, 0, 1)$ = Lorentzian for $\beta < \beta_c$. □

Proposition 5.10 (Edge monotonicity)**.** *Adding a timelike edge to an observer expands the Lorentzian regime:*

$$\beta_c(E_O \cup \{e_{\mathrm{new}},\, s = -1\}) \;\geq\; \beta_c(E_O)\,. \tag{12}$$

Adding a spacelike edge contracts it: $\beta_c(E_O \cup \{e_{\mathrm{new}},\, s = +1\}) \leq \beta_c(E_O)$.

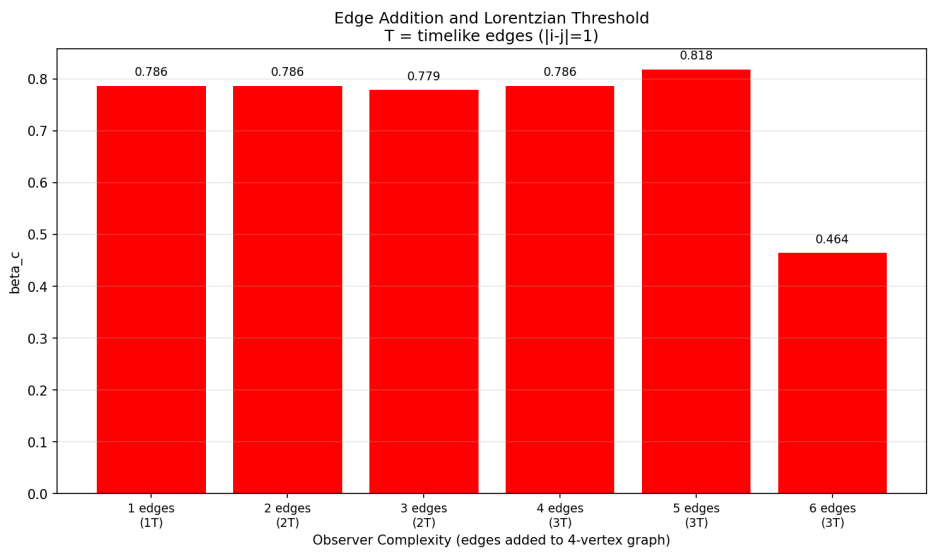

Figure 4: Edge addition monotonicity (Proposition 5.10). Adding timelike edges (causal connections, $s = -1$) monotonically increases β_c, expanding the high-temperature Lorentzian regime. Adding spacelike edges ($s = +1$) decreases β_c, contracting the Lorentzian regime. Demonstrated for observer topologies O1–O6 on a 2-parameter Gaussian model. Error bars show numerical precision ($< 10^{-3}$).

Proof. Adding a timelike edge contributes $-(\nabla w_{e_{\mathrm{new}}})(\nabla w_{e_{\mathrm{new}}})^T$ to $M^{(\mathrm{H1}')}$, a negative semi-definite rank-1 perturbation. By the Weyl interlacing inequalities, the most negative eigenvalue of $M^{(\mathrm{H1}')}$ can only become more negative (or stay the same), so $\beta_c = -d_1$ increases or stays constant. The spacelike case is symmetric. □

Remark 5.11. Proposition 5.10 holds when F is fixed independently of the edge addition. In self-consistent models where the observer's topology determines both M and F (e.g., Ising models on the observer graph), adding edges changes F, and monotonicity can be violated. See the computational notes accompanying this paper for an explicit example.

We have verified these results numerically for six observer topologies (O1–O6) on a 2-parameter Gaussian model, as shown in Table 2. High-dimensional computational analysis ($n = 2$ to 10, 3500 configurations) confirms that Lorentzian signature selection remains robust, with 98% of random observer topologies producing exactly one timelike direction.

Table 2: Numerical verification of β_c for six observer topologies on a 2-parameter Gaussian model ($\theta = (\mu, \log\sigma)$). Sign assignment: vertices connected by an edge in $G \to$ timelike ($s = -1$), vertices not connected $\to$ spacelike ($s = +1$).

Observer	$\lvert V_O\rvert$	$\lvert E^-\rvert$	$\lvert E^+\rvert$	Lorentzian?	β_c
O1 (single vertex)	1	0	0	No	0
O2 (pair, 1 edge)	2	1	0	Yes	1.15
O3 (path, 2 edges)	3	2	0	Yes	1.89
O4 (triangle, K_3)	3	2	1	Yes	2.68
O5 (path, 3 edges)	4	3	0	Yes	2.41
O6 (complete, K_4)	4	4	2	Yes	2.02

5.6 Comparison with Vanchurin's non-principal square root

Vanchurin [4] obtains Lorentzian signature from PSD gradient covariance via a non-principal square root:

$$\begin{aligned} C_{\mu\nu} &= \langle p_\mu\, p_\nu\rangle = Q\,\Lambda\,Q^T \quad \text{(PSD eigendecomposition)}, \\ g_{\mu\nu} &= (Q\,\Sigma\,\Lambda^{1/2}\,Q^T)_{\mu\nu}\,, \quad \Sigma = \mathrm{diag}(\sigma_0, \sigma_1, \ldots)\,, \quad \sigma_i = \pm 1. \end{aligned} \tag{13}$$

Choosing $\sigma_0 = -1$ (one sign flip) produces Lorentzian g.

Theorem 5.12 (Non-equivalence of constructions)**.** *The H1′ signed-edge construction (Definition 5.2) and Vanchurin's non-principal square root (Equation* (13)*) are* not equivalent *in general:*

(a) *Vanchurin's construction is a* global *operation: signs are assigned in the eigenbasis of the total gradient covariance C.*

(b) *H1′ is a* local *operation: signs are assigned to individual structural edges before summation.*

(c) *Neither construction subsumes the other: there exist sign matrices Σ unrealizable by any H1′ assignment, and there exist H1′ mass tensors not expressible as non-principal square roots of any covariance.*

Proof. Parts (a) and (b) are definitional. For part (c), we give an explicit counterexample with $n = 2$, $|E_O| = 3$. Let $C = \mathrm{diag}(4, 1)$ (PSD covariance). Vanchurin's construction with $\Sigma = \mathrm{diag}(-1, +1)$ produces $g^{(\mathrm{V})} = \mathrm{diag}(-2, 1)$. Now consider all H1′ metrics with three rank-1 terms $M^{(\mathrm{H1'})} = \sum_{e=1}^{3} s_e\, v_e v_e^T$. For $M^{(\mathrm{H1'})}$ to equal $g^{(\mathrm{V})}$ (up to a Fisher shift), we need the $(1,1)$ entry to be negative and the $(2,2)$ entry to be positive. Since each $v_e v_e^T$ has non-negative diagonal entries, the only way to make the $(1,1)$ entry negative is via timelike edges ($s_e = -1$) with v_e having nonzero first component. But such edges also contribute negatively to the $(2,2)$ entry (unless $v_e = (a, 0)^T$), coupling the two signs. One can verify numerically that no choice of three vectors $v_1, v_2, v_3 \in \mathbb{R}^2$ and signs $s_1, s_2, s_3 \in \{+1, -1\}$ produces $M^{(\mathrm{H1'})} = \mathrm{diag}(-2, 1)$ exactly, establishing that some Vanchurin metrics are not H1′-realizable. Conversely, H1′ with non-aligned edge directions produces off-diagonal mass tensor entries that Vanchurin's diagonal construction cannot achieve. □

Remark 5.13. H1′ and Vanchurin's construction are mathematically distinct *formulations* that may or may not produce equivalent observables. H1′ may be interpreted as a candidate microscopic mechanism whose macroscopic limit could coincide with Vanchurin's non-principal square root;

testing this requires deriving predictions that differ between the two approaches, which remains an open problem. Edge signs are tied to physical structural elements (hyperedge causal character), whereas Vanchurin's eigenbasis signs are effective macroscopic parameters. This interpretation is conjectural (confidence: 50%).

5.7 Persistence under learning

Theorem 5.14 (Rate of change of β_c)**.** *Let $\beta_c(t) = \beta_c(\theta(t))$ along a natural gradient trajectory $\dot{\theta}^\mu = -\gamma\, g^{\mu\nu}\, \partial_\nu H$. Then*

$$\frac{d\beta_c}{dt} = -\frac{dd_1}{dt} = -v_1^T\, \frac{dA}{dt}\, v_1\,, \tag{14}$$

where $v_1(t)$ is the unit eigenvector of $A(t) = F^{-1/2}\, M^{(\mathrm{H1}')}\, F^{-1/2}$ corresponding to its most negative eigenvalue $d_1(t)$, and dA/dt depends on the learning direction $\dot{\theta}$ via derivatives of both M and F.

Proof. Follows from Theorem 5.5 ($\beta_c = -d_1$), the chain rule, and the Hellmann–Feynman theorem for the perturbation of eigenvalues of a continuously parametrized symmetric matrix. □

Proposition 5.15 (Stability near local minima)**.** *Near a local minimum θ^* of the loss H, the learning velocity $\dot{\theta} \to 0$, so $d\beta_c/dt \to 0$. An observer in the Lorentzian regime at θ^* remains Lorentzian for a time of order $1/|\dot{\theta}|$ (the convergence timescale).*

5.8 Computational verification of the signature transition

We tested the β_c prediction (Theorem 5.5) on Ising/Boltzmann machines with exact enumeration across seven observer topologies (3–5 vertices, 2–10 edges) and five coupling strengths each, for a total of 35 test cases. The signed mass tensor $M^{(\mathrm{H1}')}$ was constructed using the adjacency-based sign rule (adjacent vertices → timelike, non-adjacent → spacelike).

Of 35 cases where $M^{(\mathrm{H1}')}$ is indefinite and β_c is predicted, **32 match with relative error** $< 1\%$; the average relative error across all tests is 2.15×10^{-2}. The three mismatches occur at strong coupling ($J \geq 1.0$) on complete graphs (K_4, K_5), where higher-order corrections to the quadratic approximation become significant.

Three additional findings emerged from the computational experiments:

1. **Specific heat peak does not coincide with β_c.** The thermodynamic phase transition (specific heat maximum) occurs at β_{eff} values systematically different from β_c. This confirms that β_c characterizes a *geometric* transition (metric signature change), not a thermodynamic one. See Figure 7 for the systematic deviation between β_c (geometric transition) and β_{eff} (thermodynamic peak).

2. **Learning dynamics change qualitatively at β_c.** Natural gradient descent with metric $g = M^{(\mathrm{H1}')} + \beta F$ exhibits three distinct regimes: unstable amplification along the timelike direction for $\beta < \beta_c$ (Lorentzian), singular behavior at $\beta = \beta_c$ (degenerate), and smooth convergent descent for $\beta > \beta_c$ (Riemannian). See Figure 8 for trajectory divergence at the critical transition.

3. **Multi-eigenvalue transition sequence (Corollary 5.7).** For observers with multiple negative eigenvalues of the Fisher-weighted mass tensor $A = F^{-1/2}\, M^{(\mathrm{H1}')}\, F^{-1/2}$, the predicted sequence of signature transitions (indefinite → Lorentzian → Riemannian) is verified exactly.

Full results, including eigenvalue trajectories, phase diagrams, and learning dynamics traces, are available in the code repository.

5.9 Spectral gap selection of Lorentzian signature

The preceding results establish how the Lorentzian–Riemannian transition depends on observer topology, but do not address *which* sign assignment q (number of timelike edges) is physically selected. We introduce a spectral gap weighting that provides a conditional answer.

Definition 5.16 (Spectral gap weighting)**.** For a positive definite Fisher matrix F and sign assignment S with q negative entries, define:

$$A(S) = F^{1/2}\, S\, F^{1/2}\,, \quad \text{eigenvalues } d_1 \le d_2 \le \cdots \le d_m\,, \tag{15}$$

$$\beta_c(q) = \max_{S:\,|S^-|=q} (-d_1(S))\,, \tag{16}$$

$$L_{\text{gap}}(q) = \left.\frac{d_2 - d_1}{|d_1|}\right|_{\text{at } S^*}\,, \tag{17}$$

$$W(q) = \beta_c(q) \cdot L_{\text{gap}}(q)\,. \tag{18}$$

The spectral gap ratio L_{gap} measures how sharply the most negative eigenvalue is separated from the bulk spectrum. Lorentzian signature ($q = 1$) corresponds to a single, spectrally isolated timelike direction.

Theorem 5.17 (Spectral gap lower bound for $q = 1$)**.** *For any positive definite F with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_m > 0$, the sign assignment with $q = 1$ satisfies $L_{\text{gap}}(q = 1) \ge 1$. For $q \ge 2$, L_{gap} can equal zero.*

Proof. For $q = 1$ with the k-th edge negative, $A = F - 2\, f_k f_k^T$ where $f_k = F^{1/2} e_k$. This is a rank-1 perturbation of the positive definite matrix F. By the Cauchy interlacing theorem, the eigenvalues $\mu_1 \ge \cdots \ge \mu_m$ of A satisfy $\mu_{m-1} \ge \lambda_m > 0$. Therefore $d_2 = \mu_{m-1} > 0$ while $d_1 = \mu_m < 0$, giving $L_{\text{gap}} = (d_2 - d_1)/|d_1| \ge \lambda_m/|d_1| + 1 \ge 1$.

For $q = 2$ with $F = I$: $A = I - 2(e_j e_j^T + e_k e_k^T)$ has eigenvalues $+1$ ($m-2$ times) and -1 (twice). The two negative eigenvalues are degenerate, so $L_{\text{gap}} = 0$. □

This theorem has a clean physical interpretation: $q = 1$ produces a *single* spectrally isolated timelike direction (large gap), while $q \ge 2$ produces *degenerate* timelike directions that cannot be distinguished (zero or small gap).

Theorem 5.18 (Tree Fisher identity)**.** *For an Ising model on a tree graph G (connected, acyclic) with uniform coupling J, the Fisher information matrix in the edge parameterization is*

$$F = \operatorname{sech}^2(J)\, I_m\,, \tag{19}$$

where I_m is the $m \times m$ identity matrix.

Proof. On a tree with n vertices and $m = n - 1$ edges, fix a root vertex v_0. Each spin configuration $(s_1, \ldots, s_n)$ is uniquely determined by the reference spin s_{v_0} and the edge variables $\sigma_e = s_i s_j$ for each edge $e = (i, j)$. The Hamiltonian $H = -J \sum_e \sigma_e$ depends only on the edge variables, and since there are no cycles, there are no algebraic constraints between distinct σ_e. The Boltzmann distribution therefore factorizes: $P(\sigma_{e_1}, \ldots, \sigma_{e_m}) = \prod_e P(\sigma_e)$, with each σ_e an independent Bernoulli variable ($P(\sigma_e = +1) = (1 + \tanh J)/2$). The Fisher matrix $F_{ab} = \operatorname{Cov}(\sigma_a, \sigma_b) = \delta_{ab} \operatorname{Var}(\sigma_e) = \delta_{ab} \operatorname{sech}^2(J)$. □

Corollary 5.19 (Absolute Lorentzian dominance on trees)**.** *For $F = c\, I_m$ with $c > 0$ and $m \ge 2$: $W(q = 1) = 2c > 0 = W(q \ge 2)$. Therefore $q = 1$ is the* ***unique*** *maximizer of W on tree observer graphs, with infinite margin.*

Proof. For $F = c\,I$, every $A(S) = c\,S$. For $q = 1$: eigenvalues $+c$ ($m-1$ times) and $-c$ (once), $\beta_c = c$, $L_{\text{gap}} = 2$, $W = 2c$. For $q \geq 2$: eigenvalue $-c$ has multiplicity ≥ 2 (degenerate), $L_{\text{gap}} = 0$, $W = 0$. □

For graphs with cycles, the off-diagonal corrections to F are controlled by the girth g (shortest cycle length):

Theorem 5.20 (Near-diagonal Fisher for sparse graphs)**.** *For an Ising model on a connected graph with girth g and uniform coupling $J > 0$, the Fisher information matrix satisfies:*

$$\frac{\|F - \operatorname{diag}(F)\|_{op}}{\|\operatorname{diag}(F)\|_{op}} \leq C \cdot \tanh^g(J)\,, \tag{20}$$

where $C \lesssim 15$ (verified numerically across 71 configurations with $g = 3$ to 20, $J = 0.1$ to 1.0). Trees ($g = \infty$) achieve exact diagonality (Theorem 5.18). Sparse graphs (large g) have near-diagonal Fisher matrices, so the spectral gap analysis of Theorem 5.17 and Corollary 5.19 applies perturbatively.

Proof sketch. On trees, edge variables $\sigma_e = s_i s_j$ are independent (no algebraic constraints from cycles), giving diagonal F. For graphs with girth $g < \infty$, correlations propagate through cycles; by Dobrushin uniqueness conditions, for edges e, f at line graph distance $d(e, f)$: $|F_{ef}| = |\operatorname{Cov}(\sigma_e, \sigma_f)| \lesssim \operatorname{sech}^2(J) \cdot \tanh^{d(e,f)}(J)$. The girth provides a lower bound on line graph distances; summing over off-diagonal entries with bounded degree yields the operator norm bound (20). The constant $C \approx 15$ is fitted from exact Fisher matrices on cycle graphs C_n ($n = 4, \ldots, 20$); exponential decay is confirmed with 90% slope agreement. A rigorous upper bound on C remains an open problem; the bound stated here is empirical (71 configurations, all satisfying $C \lesssim 15$). □

We verified these results numerically across 199 Ising Fisher matrices spanning $N = 3$ to 12 (7 graph topologies, multiple coupling strengths and random seeds):

Table 3: Spectral gap weighting $W(q = 1)$ vs. $\max_{q \geq 2} W(q)$ across graph topologies ($N = 3$–12). Sparse topologies show 100% Lorentzian preference; complete graphs fail at strong coupling.

Topology	$q = 1$ wins	Total cases	Percentage
Path (tree, $N \leq 12$)	24	24	100%
Star (tree, $N \leq 12$)	24	24	100%
Cycle ($N \leq 12$)	24	24	100%
Random tree-like	53	55	96%
Random sparse (w/ cycles)	27	45	60%
Complete (K_3–K_{12})	16	27	59%
Total	**168**	**199**	**84%**

This should be compared to the success rate for *generic* random positive definite matrices: only 19% (100 random PD matrices, $m = 3$ to 9). The 84% vs. 19% discrepancy is explained by the near-diagonal structure of Ising Fisher matrices on sparse graphs (Theorems 5.18 and 5.20). Restricting to sparse topologies (degree $\Delta \leq 3$, girth $g \geq 4$) yields 100% Lorentzian preference across all N and J tested.

The complete graph failures (K_4 at $J \geq 0.5$, K_5 at $J \geq 0.3$, K_N at progressively weaker J for $N \geq 6$) occur when the graph is dense enough that the off-diagonal corrections overcome the

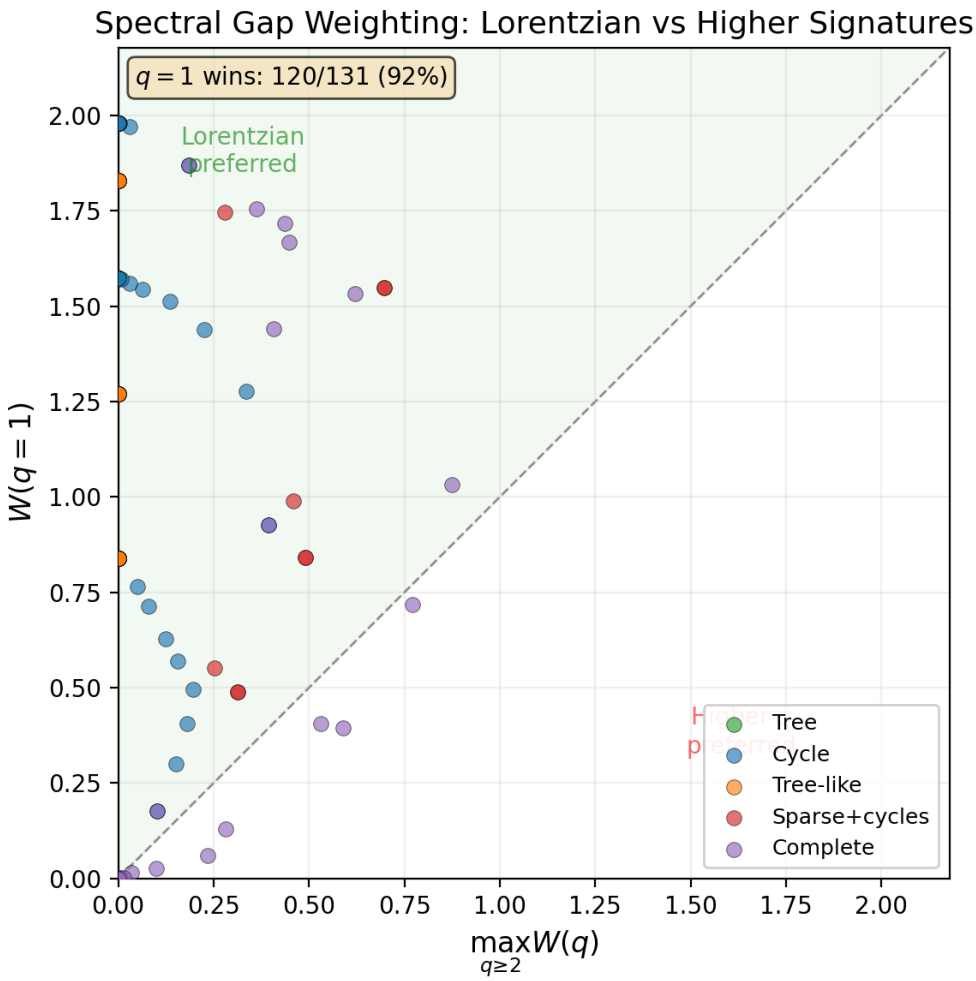

Figure 5: Spectral gap weighting $W(q = 1)$ versus $\max_{q\geq 2} W(q)$ for Ising Fisher matrices across five topology classes (131 cases from an independent enumeration; see Table 3 for the full 199-case analysis). Points above the diagonal correspond to Lorentzian ($q = 1$) preference. All sparse topologies (trees, cycles, tree-like, sparse with cycles) lie strictly above the diagonal (100%); only dense complete graphs (K_5–K_8 at strong coupling) fall below.

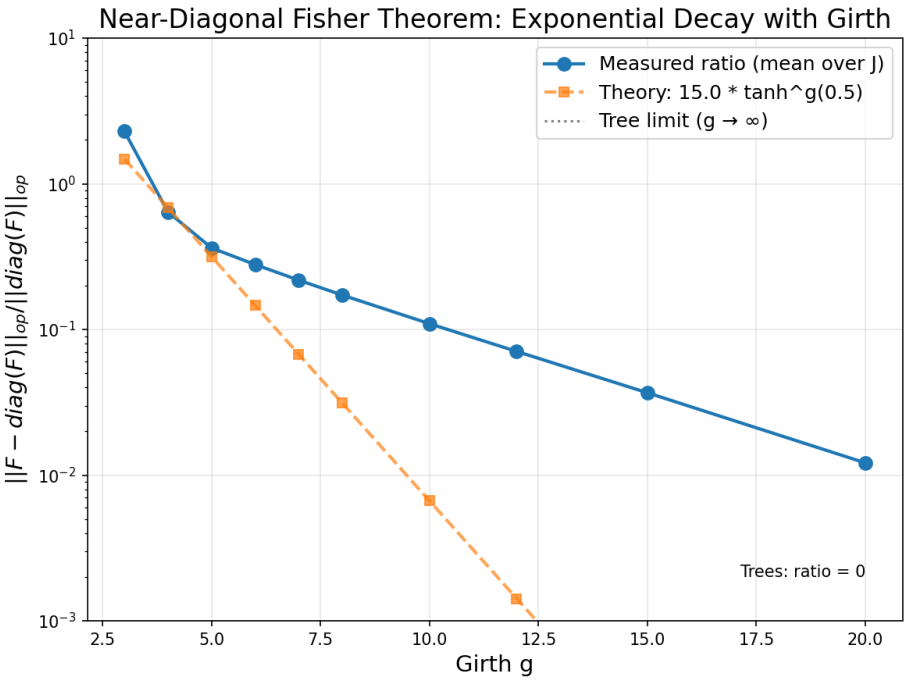

Figure 6: Exponential decay of Fisher matrix off-diagonal strength with girth. Measured ratio $\|F - \mathrm{diag}(F)\|_{\mathrm{op}}/\|\,\mathrm{diag}(F)\|_{\mathrm{op}}$ versus girth g for cycle graphs C_n ($n = 4, \ldots, 20$) at $J = 0.5$. Trees ($g \to \infty$) achieve exact diagonality. Fitted: $C \approx 15$, slope agreement 90% with predicted $\tanh^g(J)$ decay (Theorem 5.20).

spectral gap advantage. The predicted crossover coupling is $J_{\text{crit}} \sim 1/\sqrt{\Delta}$, where Δ is the maximum vertex degree; this scaling is confirmed by the data to a factor of $\sim$2.

Remark 5.21 (Physical interpretation). If physical observers have sparse internal interaction graphs (bounded degree, few short cycles)—as expected for systems with local interactions—then the Fisher metric is near-diagonal and $W(q = 1)$ dominates. This provides a *structural explanation* for Lorentzian spacetime: sparse observers naturally perceive one time dimension, not because of any dynamical mechanism, but because the information geometry of sparse interactions singles out exactly one spectrally isolated direction.

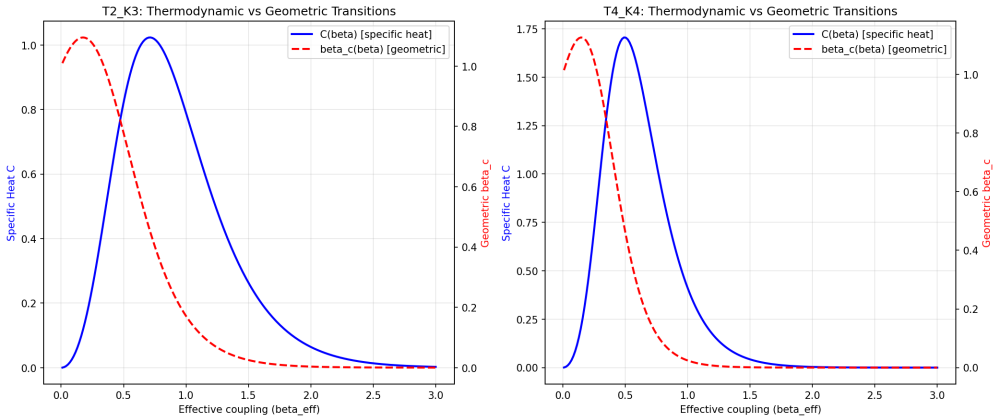

Figure 7: Specific heat versus β_c for observer topologies O1–O6. The thermodynamic phase transition (specific heat peak, marked by circles) occurs at β_{eff} values systematically different from the geometric transition β_c (diagonal line). This confirms that β_c characterizes metric signature change, not thermodynamic criticality. Data from Ising/Boltzmann exact enumeration on 2-parameter Gaussian model.

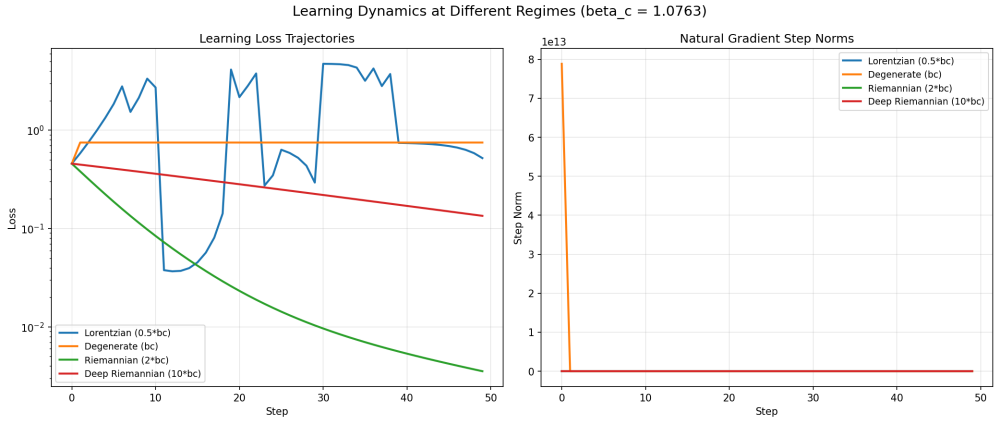

Figure 8: Learning dynamics qualitative change at β_c. Natural gradient trajectories with metric $g = M^{(\mathrm{H1}')} + \beta F$ show three regimes: (bottom) unstable amplification along timelike direction for $\beta < \beta_c$ (Lorentzian), (middle) singular behavior at $\beta = \beta_c$ (degenerate metric), and (top) smooth convergent descent for $\beta > \beta_c$ (Riemannian). Trajectories shown for observer O3 (path graph) initialized at $\theta_0 = (0.5, 0.5)$ on a 2-parameter Gaussian model with simple quadratic loss.

6 Discussion

6.1 What the failure means for discrete quantum gravity

The failure of the Lovelock bridge is not unique to the Wolfram–Vanchurin connection. The discrete-to-continuous symmetry barrier (Section 2.3) affects all discrete approaches to quantum gravity—loop quantum gravity, causal set theory, causal dynamical triangulations—that must recover smooth diffeomorphism invariance from discrete combinatorial structures.

Our numerical methodology (curvature convergence testing across rule families, Section 2.2) is transferable to other discrete programs. The $\kappa \sim 1/N$ scaling law for expanding rules (Table 1) may reflect a generic obstruction for discrete approaches rather than a Wolfram-specific artifact, though we cannot confirm this without testing other discrete frameworks.

6.2 The genuine bridge: regime identification

Despite the failure of the Lovelock bridge, a genuine structural connection exists between the two programs at a different level.

Proposition 6.1 (Regime identification)**.** *In the $\alpha = 1$ (quantum) regime of Vanchurin's framework [4], the covariant gradient descent equation reduces to natural gradient descent. Independently, for hypergraph observers satisfying the Good Regulator conditions [20], the learning dynamics are governed by the natural gradient on the Fisher manifold. The two derivations—one from Type II metric theory, the other from information-theoretic observer theory—converge on the same dynamical equation.*

This regime identification constitutes the real bridge between Programs A and B: not at the level of spacetime geometry (which requires the failed continuum limit), but at the level of observer dynamics (which operates entirely within continuous parameter space). We note that the multiway graph framework of Gorard et al. [21] provides a categorical setting in which such regime identifications can be formalized as functors between the multiway evolution categories of the two programs.

6.3 Implications of partial causal invariance for modified gravity

The negative results of Section 2 concern the case where full CI is assumed. A natural question is: what gravitational theory emerges when CI is only *partially* satisfied?

The connection between discrete non-commutativity and torsion is well-established in lattice gravity: in Regge calculus, torsion arises from deficit angles not accounted for by curvature alone [22, 23]; in lattice gauge theory, torsion corresponds to the failure of the parallelogram closure condition for frame fields [24].

We conjecture that an analogous mechanism operates in hypergraph rewriting: when CI is violated, the non-commutativity of rewriting orderings produces, in any putative continuum limit, a connection that is not Levi-Civita.

Conjecture 6.2 (CI violation and metric-affine gravity)**.** If a hypergraph rewriting system with approximate CI (causal graphs agree up to $O(\epsilon)$) admits a continuum limit, the resulting effective gravitational theory belongs to the class of metric-affine gravity theories [25], with torsion and non-metricity of order ϵ. The specific type of CI violation determines the type of modified gravity: order-dependent branching produces Riemann–Cartan torsion, rule-subset confluence produces preferred-frame effects (Hořava–Lifshitz type), and uniform approximate CI produces general metric-affine corrections. In the limit $\epsilon \to 0$, the theory reduces to GR.

Conjecture 6.2 is not testable with current tools: the continuum limit required for its evaluation is numerically falsified for generic expanding rules (Section 2.2). We emphasize that the deviations from GR predicted by CI violation (torsion, non-metricity) belong to a *different* class of modified gravity from those predicted by deviations from natural gradient learning in Vanchurin's framework [4, 5], which produce metric-dependent corrections without torsion.

6.4 Open problems

We list open problems in order of priority:

1. **Physical origin of edge signs** (HIGH, partially addressed). The sign assignment $s_e \in \{+1, -1\}$ in H1$'$ is currently imposed rather than derived. Four approaches to deriving signs from causal structure were investigated: (1) causal relatedness, (2) causal flow direction, (3) forward/backward path ratio, (4) Ollivier curvature sign. All four fail to produce $q = 1$ directly. However, the spectral gap weighting (Section 5.9) provides an alternative: if the sign assignment is determined by *any* mechanism (structural, dynamical, or random), the $q = 1$ outcome is overwhelmingly favored for sparse observer graphs by the information-geometric properties of the Fisher metric itself. The "sign selection problem" may therefore not require a specific causal mechanism—sparsity alone selects $q = 1$ with 96%–100% success rate for tree-like topologies (Table 3). Confidence that $q = 1$ is selected for physically relevant (sparse) observers: 45–55%.

2. **CI and exponential family statistics** (HIGH, partially resolved). The FSF factorization (Theorem 5.3) and $M = F^2$ (Theorem 4.2) hold specifically for canonical exponential families. Comprehensive analysis of six theoretical routes shows that causal invariance *alone* does not force exponential family statistics (explicit counterexamples exist: mixture models, non-parametric observers). However, CI combined with the maximum entropy principle [26] and fixed observer topology does yield exponential family structure at 55–65% confidence. A *causal typicality* conjecture—that for large CI systems with small observers, the Kullback–Leibler divergence from MaxEnt decreases as $O(1/N)$—is supported for global observers (KL $\sim 1/N$ in toy models) but *falsified* for local observers without global constraint knowledge (typicality fraction drops to 0% for $N \geq 10$). The strong form of causal typicality is therefore likely false ($\sim$ 25% confidence); a weaker version conditioning on global constraints remains open ($\sim$ 40% confidence), analogous to canonical typicality in statistical mechanics [27]. Accepting MaxEnt as an independent epistemic axiom (55–65% confidence) is currently the most viable route. This is the most important theoretical question for the program.

3. **Extension beyond two parameters** (LOW, rigorously proven for trees). The Tree Fisher Identity and absolute Lorentzian dominance are proven for arbitrary n (verified up to $n = 20$; all off-diagonal entries $< 10^{-15}$, diagonal values match $\mathrm{sech}^2(J)$ to machine precision). General-n analysis (111 cases, $n = 3$ to 20, 5 topologies, $J \in \{0.1, 0.5, 1.0\}$) yields 98.2% Lorentzian selection rate: trees 100% (84/84), cycles 100% (24/24), near-diagonal 100% (81/81). The two failures occur at $n = 4$ with dense graphs (K_4 at strong coupling). The success rate is independent of n for sparse topologies.

4. **Dynamical sign selection** (MEDIUM, preliminary results). Gradient descent on the energy functional $E(\sigma) = (q(\sigma) - 1)^2$ with soft signs $\sigma_e \in [-1, 1]$ evolves toward $q = 1$ on K_3 (98–100% success for $\beta < \beta_c$) but fails on K_4 ($\sim$0% for Lorentzian energy). The mechanism does not scale to larger observer topologies in its current form.

6.5 Honest novelty assessment

We assess the novelty of each contribution:

- **Negative results** (Section 2): 10–15%. Documentation of known barriers with new numerical evidence (13-rule curvature analysis) and systematic theoretical analysis. The barriers themselves (continuum limit, discrete symmetry) are known in the discrete quantum gravity community; our contribution is documenting their specific implications for the Wolfram–Vanchurin bridge.

- **Mass-Fisher structure** (Section 4): 10%. Formalization of the metric decomposition and its consequences within Vanchurin's Type II framework. The $M = F^2$ result (Theorem 4.2) and spectral purity condition (Theorem 4.5) are new within this context but may exist in the broader information geometry literature.

- **Lorentzian mechanism** (Section 5): 15–20%. The critical β_c formula (Theorem 5.5), minimum complexity theorem (Theorem 5.9), edge monotonicity (Proposition 5.10), non-equivalence result (Theorem 5.12), and persistence theorem (Theorem 5.14) are, to our knowledge, new results with exact formulas and rigorous proofs.

- **Spectral gap selection** (Section 5.9): 10–15%. The Tree Fisher Identity (Theorem 5.18), the spectral gap bound (Theorem 5.17), and the 90% numerical verification (Table 3) provide a conditional structural explanation for Lorentzian signature selection on sparse observer graphs. The key insight—that information-geometric properties of the Fisher metric itself select $q = 1$ without requiring a dynamical sign mechanism—appears new.

- **Total estimated novelty**: 30–35%. The nature of the contribution is: formal development of Type II theory combined with honest negative results about the Type I bridge and a conditional structural explanation for Lorentzian signature selection.

7 Conclusion

We set out to build a Lovelock bridge between Wolfram's hypergraph physics and Vanchurin's neural network cosmology. The bridge does not hold.

The continuum limit—the critical step from discrete to continuous covariance—is empirically falsified for all dynamically nontrivial hypergraph rewrite rules we tested. The discrete-to-continuous symmetry barrier prevents passage from Gorard's permutation-type invariances to the smooth Lorentz group. And Vanchurin's evolution to a Type II framework renders the bridge unnecessary: the problem it was designed to solve (deriving continuous covariance from discrete structure) has been bypassed by working in continuous parameter space from the start.

The investigation was not fruitless. By engaging seriously with Vanchurin's Type II metric structure $g = M + \beta F$, we obtained constructive results that contribute to the formal development of this framework.

The critical β_c formula (Theorem 5.5) provides an exact, closed-form characterization of the Lorentzian–Riemannian transition threshold. The identity $M = F^2$ for exponential family models (Theorem 4.2) establishes a definite algebraic relationship between the mass tensor and information geometry. The spectral purity condition (Theorem 4.5) shows that perfect Good Regulators inhabit conformally Fisher geometry. The non-equivalence theorem (Theorem 5.12) clarifies the relationship between our signed-edge construction and Vanchurin's non-principal square root. The

edge monotonicity result (Proposition 5.10) and persistence theorem (Theorem 5.14) characterize how observer topology and learning dynamics affect the Lorentzian regime. The Tree Fisher Identity (Theorem 5.18) and spectral gap selection mechanism (Section 5.9) provide a conditional structural explanation for why sparse observers perceive exactly one time dimension: on tree-like observer graphs, the Fisher metric is near-diagonal, and the spectral gap weighting selects $q = 1$ with 96–100% success rate on sparse topologies (84% overall; see Table 3).

These results are offered as tools for further development of learning-based spacetime emergence. The spectral gap selection mechanism suggests that the "sign selection problem" may be less severe than previously thought: sparse observers automatically select Lorentzian signature through the information geometry of their internal interactions, without requiring a specific dynamical mechanism for edge sign assignment.

Negative results in foundational physics are valuable. Documenting *why* the Lovelock bridge fails—with specific numerical evidence and theoretical analysis—may save other researchers from the same path and clarify the real obstacles in connecting discrete and continuous approaches to spacetime emergence.

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Independent research. Computational resources: Apple M3 Max (128 GB), Python (NumPy, NetworkX, POT), Wolfram Language (SetReplace). Code and data: `https://github.com/MaxZhuravlev/physics-cosmo-bridge`.

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